

Lecture 0

Prerequisites: A Refresher

This sheet begins with a short diagnostic of the four computational skills the early lectures rely on. Take only the seven problems listed below *cold*, without notes, and stop after 45 minutes. Then check your work and use the routing guide at the end: repair only the weak strand and complete its recheck. The remaining algebra problems form a targeted bank, not a second cold test. Throughout, $\mathbb{F}$ denotes $\mathbb{R}$ or $\mathbb{C}$.

Scored algebra diagnostic (40–45 minutes). Attempt problems **0.1, 0.2, 0.9, 0.15, 0.22, 0.23, and 0.30**. Award one point only for a correct result supported by enough working to identify the method. These seven checks cover complex arithmetic and polar form, compatible matrix operations, determinants, row reduction, null-space bases, and consistency-first classification.

- **Ready:** at least 6/7, including 0.9, 0.15, and 0.30, plus at least one correct complex check (0.1–0.2) and one correct row-reduction check (0.22–0.23).
- **Targeted repair:** 4–5 points, or any missing required strand above. Use the matching refresher section and bank route, then complete the named recheck before Lecture 1.
- **Rebuild first:** 0–3 points, or gaps in two or more strands. Work through each weak route with notes open, then retake the seven diagnostic checks on a later day.

Separate analysis-readiness route. Problems 0.32–0.36 are a non-scored, 15–20 minute routing check for differentiation, integration by parts, elementary linear ODEs, series, and Fourier coefficients. They are not part of the seven-point algebra score and are not assumed in Lecture 0 itself. Use them before the later application weeks; this sheet is not a replacement calculus or analysis course.

Complex arithmetic

Exercise 0.1 (diagnostic) Let $z = 3 + 2i$ and $w = 1 + i$. Compute $z + w$, zw , z/w , the conjugate $\bar{z}$, and the modulus $|z|$.

Exercise 0.2 (diagnostic) Write $1 - i$ in polar form and use it to compute $(1 - i)^8$.

Exercise 0.3 (computational) Find all complex numbers z with $z^2 = i$.

Exercise 0.4 (computational) Let $z = 3 + i$ and $w = 1 + 2i$. Compute $z + w$, zw , z/w , the conjugate $\bar{z}$, and the modulus $|z|$.

Exercise 0.5 (computational) Write $1 + i$ in polar form and use it to compute $(1 + i)^6$.

Exercise 0.6 (computational) Write $\sqrt{3} + i$ in polar form and use it to compute $(\sqrt{3} + i)^4$, giving the answer in Cartesian form.

Exercise 0.7 (bridge) For complex numbers $z = a + bi$ and $w = c + di$, prove that $\overline{zw} = \bar{z} \bar{w}$.

Exercise 0.8 (extension) Find all four complex numbers z with $z^4 = -4$.

Matrix algebra

Exercise 0.9 (diagnostic) Let $A = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$. Compute AB , BA , and A^T , and verify $\text{tr}(AB) = \text{tr}(BA)$ even though $AB \neq BA$.

Exercise 0.10 (computational) Let $A = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix}$. Compute AB and BA , and verify $\text{tr}(AB) = \text{tr}(BA)$ although $AB \neq BA$.

Exercise 0.11 (computational) For $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$, compute A^2 and A^3 .

Exercise 0.12 (bridge) Let $A = \begin{pmatrix} 1+i & 2-i \\ 0 & 3i \end{pmatrix}$ over $\mathbb{C}$. Write down the transpose A^T and the conjugate transpose $A^\dagger = \overline{A}^T$.

Exercise 0.13 (extension) For square matrices A, B of the same size, prove the cyclic identity $\text{tr}(AB) = \text{tr}(BA)$.

Exercise 0.14 (bridge) For matrices $A \in \mathcal{M}_{m \times n}(\mathbb{F})$ and $B \in \mathcal{M}_{n \times p}(\mathbb{F})$, prove that $(AB)^T = B^T A^T$.

Determinants

Exercise 0.15 (diagnostic) Evaluate $\det \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix}$ and state whether the matrix is invertible.

Exercise 0.16 (computational) Use the 2×2 formula to invert $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

Exercise 0.17 (bridge) For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, find all λ for which $\det(\lambda I - A) = 0$. (These values reappear in Week 5 as the eigenvalues of A .)

Exercise 0.18 (computational) Evaluate $\det \begin{pmatrix} 1 & 2 & 1 \\ 0 & 3 & 2 \\ 1 & 0 & 1 \end{pmatrix}$ and state whether the matrix is invertible.

Exercise 0.19 (computational) Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$. Compute $\det A$, $\det B$, and $\det(AB)$, and confirm $\det(AB) = \det A \det B$.

Exercise 0.20 (extension) Evaluate the block-diagonal determinant $\det \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 1 & 3 \end{pmatrix}$.

Exercise 0.21 (extension) For 2×2 matrices A, B , prove that $\det(AB) = \det A \det B$ directly from the entries.

Gaussian elimination

Exercise 0.22 (diagnostic) Solve, by row reduction,

$$\begin{aligned}x + y + z &= 2, \\x - y + 2z &= 5, \\2x + y - z &= 2.\end{aligned}$$

Exercise 0.23 (diagnostic) Find a basis for the solution space (null space) of $Ax = 0$, where $A = \begin{pmatrix} 1 & -1 & 2 & 0 \\ 2 & -2 & 3 & 1 \end{pmatrix}$, and state its dimension.

Exercise 0.24 (computational) Compute A^{-1} by row-reducing the augmented block $[A \mid I]$, for $A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$.

Exercise 0.25 (computational) Solve, by row reduction,

$$\begin{aligned}x + y + z &= 6, \\x - y + z &= 2, \\2x + y - z &= 1.\end{aligned}$$

Exercise 0.26 (computational) Describe the full solution set of

$$\begin{aligned}x + 2y - z &= 3, \\2x + y + z &= 3,\end{aligned}$$

in the form “particular solution + span of the null space”.

Exercise 0.27 (computational) Compute A^{-1} by row-reducing the augmented block $[A \mid I]$, for $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$.

Exercise 0.28 (bridge) Let $A \in \mathcal{M}_{m \times n}(\mathbb{F})$. Prove that the null space $\text{null } A = \{x \in \mathbb{F}^n : Ax = 0\}$ is closed under addition and scalar multiplication (so it is a subspace of $\mathbb{F}^n$).

Exercise 0.29 (extension) For which real values of a does the system

$$\begin{aligned}ax + y + z &= 1, \\x + ay + z &= 1, \\x + y + az &= 1\end{aligned}$$

have a unique solution, infinitely many solutions, or no solution? Give the solution in the unique case.

Exercise 0.30 (diagnostic) Classify each displayed RREF augmented matrix as having no solution, a unique solution, or infinitely many solutions. For each consistent system, give its full solution set.

$$(a) \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -1 \end{array} \right], \quad (b) \left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 0 & 1 \end{array} \right], \quad (c) \left[\begin{array}{ccc|c} 1 & 0 & -2 & 4 \\ 0 & 1 & 3 & -1 \end{array} \right].$$

Exercise 0.31 (repair) Row-reduce and classify each system, giving the full solution set when it is consistent:

- (a) $x + y = 1, \quad x - y = 3,$
- (b) $x + y = 1, \quad 2x + 2y = 3,$
- (c) $x + y = 1, \quad 2x + 2y = 2.$

Separate analysis-readiness route

Take these five checks without notes in 15–20 minutes before the application weeks. They do not contribute to the algebra-readiness score.

Exercise 0.32 (analysis route) Differentiate $f(t) = t^2 e^{-t}$.

Exercise 0.33 (analysis route) Use integration by parts to evaluate $\int_0^1 t e^t dt$.

Exercise 0.34 (analysis route) Find the general real solution of $y'' - 3y' + 2y = 0$.

Exercise 0.35 (analysis route) Determine whether $\sum_{n=0}^{\infty} 3^{-n}$ converges and, if it does, find its sum.

Exercise 0.36 (analysis route) For $f(x) = x$ on $[-\pi, \pi]$, compute the first Fourier sine coefficient $b_1 = \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin x dx$.

Reading your results

Follow only the route for a missed diagnostic strand. Work with the refresher open, close it for the named recheck, and require a correct result with working before declaring the strand repaired.

- ▶ **Complex arithmetic** (diagnostic 0.1–0.2): revisit *Refresher* §1; practise 0.3–0.6; recheck with 0.4 and 0.5. Problem 0.7 is a proof bridge and 0.8 is extension, neither scored for readiness.
- ▶ **Matrix algebra** (diagnostic 0.9): revisit *Refresher* §2; practise and recheck with 0.10–0.11. Problem 0.12 is an adjoint bridge and 0.13–0.14 are proof extensions/bridges, excluded from readiness.
- ▶ **Determinants** (diagnostic 0.15): revisit *Refresher* §3; practise 0.16 and 0.18–0.19; recheck with 0.18. Problem 0.17 is the Week 5 characteristic-polynomial bridge; 0.20–0.21 are extensions.
- ▶ **Gaussian elimination** (diagnostic 0.22–0.23 and 0.30): revisit *Refresher* §4; practise 0.24–0.27; recheck consistency and full solution sets with 0.31. Problem 0.28 is a Lecture 1 subspace bridge and 0.29 is a parameter-family extension, neither scored for readiness.

Analysis routing. For problems 0.32–0.36, 4–5 correct means the later application bridge is ready; 2–3 means repair only the named topics; and 0–1 means arrange a calculus/methods review before those weeks. Route 0.32 to product/chain rules, 0.33 to integration by parts, 0.34 to constant-coefficient ODEs (before Lectures 1 and 6), 0.35 to geometric series (before Lectures 6 and 13), and 0.36 to Fourier coefficients (before Lecture 13). Use prior calculus/methods notes or tutorial support, work two similar examples, then redo the missed check without notes on a later day.